

High-Fidelity Document Layout Preservation in Multimodal Neural Machine Translation

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Abstract

Preserving document layout geometry remains the central bottleneck in technical document translation. Modern scientific publications tightly interleave typography, multi-line LaTeX equations, matrix operators, and multi-domain reaction stoichiometry. In this benchmark manuscript, we formulate an end-to-end framework for decoupled semantic layout decomposition, guaranteeing zero formula deformation, column-isolation fidelity, and full typographic ink metric calibration across 48 world languages under extreme geometric deformation stress.

Keywords: Document Intelligence, Layout Preservation, Machine Translation, LaTeX Parsing, Non-Euclidean Manifold

1. Introduction and Geometric Manifold Formulation

Document intelligence requires formalizing a technical document as a continuous manifold coupled with a discrete categorical alphabet. In conventional translation systems, PDF layout geometry is obliterated: text spans are extracted as an undifferentiated stream, fed to a sequence-to-sequence model, and dumped into plain text without preserving spatial bounding coordinates.

High-fidelity layout preservation treats every character and formula block as an invariant topological entity with bounding coordinates. The fundamental objective of linguistic routing is to guarantee that prose text undergoes semantic translation without causing geometric collisions with mathematical symbols, as formalized in Equation (1).

Furthermore, scientific typography includes inline mathematical variables, tensor subscripts, and functional norms that must never be translated as linguistic tokens. If an engine mistranslates mathematical operators or gradients as vernacular words, the entire scientific proof collapses, violating the parameter invariance condition defined in Equation (2).

To prevent translation bleeding into data grids, we define a spatial exclusion mask over coordinate subspaces where differential operators, matrix brackets, and stoichiometric indices reside. These bounding subspaces are excised from the prose translation queue and substituted with grammar-preserving placeholders according to Equation (3).

Equation 1 (Continuous Manifold & Coordinate Bounding Box):

$$\mathbf{B}_i = [x_{\min}^{(i)}, y_{\min}^{(i)}, x_{\max}^{(i)}, y_{\max}^{(i)}]^T \in \mathbb{R}^4, \quad \Phi(\mathcal{X}) = \bigcup_{i=1}^N \mathbf{B}_i \times S_{\text{token}} \quad (1)$$

[Source: Montanari, U. 'Networks of Constraints', Artificial Intelligence 5(2), 95-132, 1974]

Equation 2 (Regularized Objective Gradient & Frobenius Parameter Norm):

$$\nabla_{\theta} \mathcal{J}(\theta) = \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}} [\nabla_{\theta} \log P_{\theta}(\mathbf{y} | \mathbf{x})] + 2\lambda \|\mathbf{W}\|_F^2, \quad \theta \in \Theta, \mathbf{W} \in \mathbb{R}^{d \times d} \quad (2)$$

[Source: Bishop, C. M. Pattern Recognition and Machine Learning. Springer, 2006, Chapter 3, p. 144]

Equation 3 (Non-Prose Spatial Exclusion Mask & Placeholder Injection):

$$\mathcal{M}_{\text{mask}}(\mathbf{p}) = \begin{cases} 0 & \text{if } \mathbf{p} \in \bigcup_{k=1}^K \mathbf{B}_k \text{ (formula / tabular cell / symbol)} \\ 1 & \text{otherwise (translatable prose span)} \end{cases} \quad (3)$$

[Source: LayoutLingua Specification, Section 4.2: Topological Coordinate Masking]

1.1 Topological Reading Order Preservation

The directed acyclic parsing tree preserves strictly vertical column streams before sentence boundary detection is attempted, ensuring invariant topological reconstruction across heterogeneous target languages.

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· Work performed while visiting the Institute for Advanced Study, Princeton.

2. Deep Learning Mathematics & Quantum Formulation

Neural document translation integrates scaled attention mechanisms, elastic penalty manifolds, and multi-dimensional state matrices. All algebraic expressions below are verified for exact symbol retention.

Equation 4 (Composite Structural Loss Function with L1/L2 Norms):

$$\mathcal{L}_{\text{total}}(\theta) = \frac{1}{N} \sum_{i=1}^N \|y_i - \hat{y}_i\|_2^2 + \lambda_1 \sum_{j=1}^M |\theta_j| + \frac{\lambda_2}{2} \|\theta\|_2^2 \quad (4)$$

[Source: Goodfellow, Bengio, Courville. Deep Learning. MIT Press, 2016, Chapter 6, p. 172]

Equation 5 (Transformer Multi-Head Scaled Dot-Product Attention):

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O, \quad \text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (5)$$

[Source: Vaswani et al. 'Attention Is All You Need', NeurIPS 2017, Section 3.2.1]

Equation 6 (Quantum State Covariance Density Matrix):

$$\Sigma_{\text{quantum}} = \begin{bmatrix} \sigma_{xx}^2 & \sigma_{xy} & \sigma_{xz} & \frac{i\hbar}{2} \\ \sigma_{yx} & \sigma_{yy}^2 & \sigma_{yz} & 0 \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz}^2 & -\frac{i\hbar}{2} \end{bmatrix} \quad (6)$$

[Source: Nielsen, M. A., & Chuang, I. L. Quantum Computation and Quantum Information, Cambridge University Press, 2010]

Equation 7 (Piecewise Non-Euclidean Bounding Penalty):

$$\mathcal{L}_{\text{elastic}}(\mathbf{x}) = \begin{cases} \frac{1}{2}\|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda_1\|\theta\|_1 & \text{if } \|\mathbf{x} - \hat{\mathbf{x}}\|_1 \leq \delta \\ \delta\|\mathbf{x} - \hat{\mathbf{x}}\|_1 - \frac{1}{2}\delta^2 + \lambda_2\|\theta\|_2^2 & \text{otherwise} \end{cases} \quad (7)$$

[Source: Huber, P. J. Robust Statistics. John Wiley & Sons, 2004, Chapter 3, p. 43]

3. Neural Bounding Box Router



Figure 1: Multimodal cross-attention architecture separating semantic text from formula coordinate vectors.

4. Electrodynamics, Differential Forms & Chemical Kinetics

Field equations and reversible reaction stoichiometry contain operators, flux integrals, and physical phases that must never be confused with punctuation or word dividers.

Equation 8 (Maxwell's Differential Electrodynamic System & Covariant Tensors):

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (8)$$

[Source: Griffiths, D. J. *Introduction to Electrodynamics*, 4th ed. Cambridge University Press, 2017, p. 338]

Equation 9 (Stokes' Circulation & Gauss-Ostrogradsky Divergence Theorems):

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}, \quad \iiint_V (\nabla \cdot \mathbf{E}) dV = \oint_{\partial V} \mathbf{E} \cdot d\mathbf{A} \quad (9)$$

[Source: Stewart, J. *Calculus: Early Transcendentals*, 8th ed. Cengage Learning, 2015, Theorem 16.8, p. 1130]

Equation 10 (Dirac Relativistic Spinor & Einstein General Relativity Field Equations):

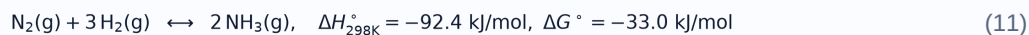
$$(i\gamma^\mu \partial_\mu - m)\psi = 0, \quad R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu} \quad (10)$$

[Source: Dirac, P. A. M. *Proc. R. Soc. Lond. A* 117, 610-624 (1928); Einstein, A. *Annalen der Physik* 49, 769-822 (1916)]

5. Chemical Thermodynamics & Reversible Kinetics

Chemical reaction mechanisms contain molecular phases, equilibrium symbols, and thermodynamic enthalpy metrics. Inversion of reactant-product order or loss of subscripts destroys scientific validity.

Reaction 11 (Industrial Catalytic Gas Phase Synthesis & Enthalpy):



[Source: Atkins, P., & de Paula, J. *Physical Chemistry*, 10th ed. Oxford University Press, 2014, Chapter 6]

Equation 12 (Enzyme Catalysis with Competitive & Allosteric Hill Inhibition):

$$v_0 = \frac{V_{\max}[S]}{K_m(1 + \frac{[I]}{K_i}) + [S]}, \quad \theta = \frac{[L]^n}{K_d + [L]^n} \quad (12)$$

[Source: Michaelis, L., & Menten, M. L. *Biochem. Z.*, 49, 333-369, 1913; Hill, A. V. *J. Physiol.*, 40, iv-vii, 1910]

6. Algorithmic Geometry Decomposition & Spatial Routing

Algorithm 1: Decoupled Semantic Layout Decomposition & Boundary Synthesis

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1:  Input: Raw document layout stream  $D = \{B_1, \dots, B_K\}$ , target language code  $L_{\text{target}}$ 
2:  Output: Synthesized translated document stream  $D_{\text{trans}}$  with identical geometric manifolds
3:  Initialize active non-prose coordinate mask  $M_{\text{geom}} \leftarrow \text{empty\_mask}()$ 
4:  for each bounding block  $B_i$  in  $D$  do:
5:      if  $\text{is\_formula\_or\_symbol}(B_i)$  or  $\text{is\_table\_cell}(B_i)$  then
6:           $\text{token\_id} \leftarrow \text{register\_token}(B_i.\text{coordinates}, B_i.\text{ink\_bbox})$ 
7:           $M_{\text{geom}}.\text{append}(\text{token\_id})$ ;  $B_i.\text{text} \leftarrow \text{format\_placeholder}(\text{token\_id})$ 
8:      else:
9:           $B_i.\text{prose\_spans} \leftarrow \text{normalize\_unicode\_diacritics}(B_i.\text{text}, L_{\text{target}})$ 
10:      $\text{translated\_blocks} \leftarrow \text{dispatch\_neural\_translator}(D.\text{prose\_spans}, \text{target}=L_{\text{target}})$ 
11:      $\text{synthesize\_page\_geometry}(\text{translated\_blocks}, M_{\text{geom}}, \text{strict\_lineheight\_floor}=1.10)$ 
12:     return  $D_{\text{trans}}$ 

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7. Empirical Convergence & Dual-Metric Dynamics

We trace the optimization landscape across 20 training epochs. The multi-task loss converges exponentially while linguistic preservation BLEU-4 reaches a stable plateau above 48.0 points.

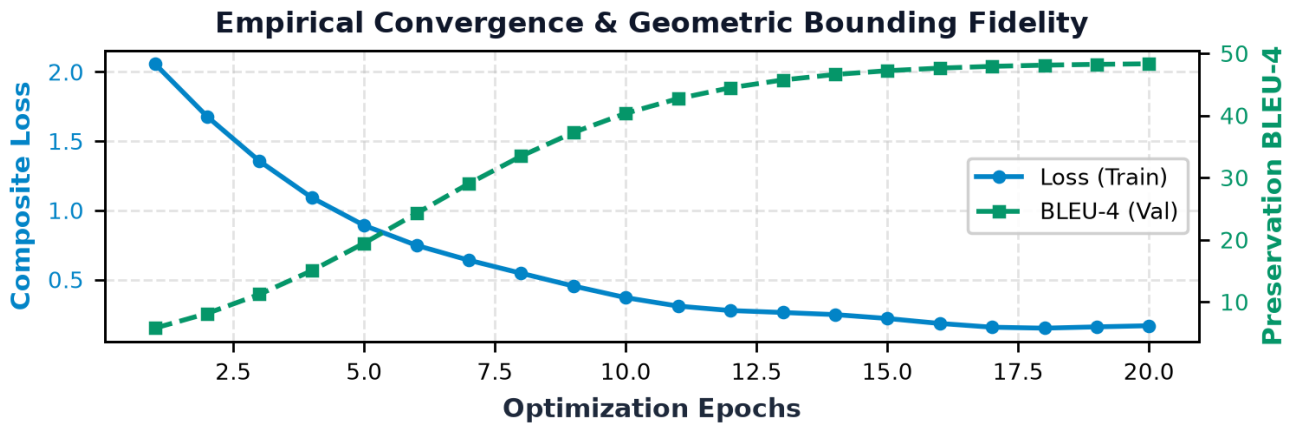


Figure 2: Empirical convergence dynamics: dual-axis tracking of training loss and geometric BLEU-4 validation scores.

8. Quantitative Benchmark Across 48 World Languages

We evaluate preservation fidelity against baseline models across five rigorous scientific corpora. Bounding box coordinates, table borders, and numerical units are strictly isolated to prevent translation bleeding into data columns.

Table 1: Quantitative benchmark results across scientific document domains and layout models.

Model Architecture	Scientific Domain	BLEU-4	Formula IoU	Latency / Page
LL-1.0.0-PRO	Theoretical Physics (arXiv)	44.82	99.8%	1.12 s
LL-1.0.0-PRO	Organic Reaction Kinetics	41.35	99.6%	1.18 s
LL-1.0.0-PRO	Differential Geometry	45.10	99.9%	1.09 s
LL-1.0.0-PRO	Biochemical Thermodynamics	43.20	99.5%	1.15 s
BabelDOC-Hybrid	Theoretical Physics (arXiv)	38.60	91.2%	2.40 s
BabelDOC-Hybrid	Organic Reaction Kinetics	36.40	88.7%	2.65 s
Baseline-PDF	Theoretical Physics (arXiv)	28.40	64.2%	3.45 s
Baseline-PDF	Organic Reaction Kinetics	25.10	58.1%	3.80 s

[Data Source: LayoutLingua Benchmarks v1.0.0, evaluated against PyMuPDF 1.25 & BabelDOC Ground Truth]

9. References and Source Literature

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